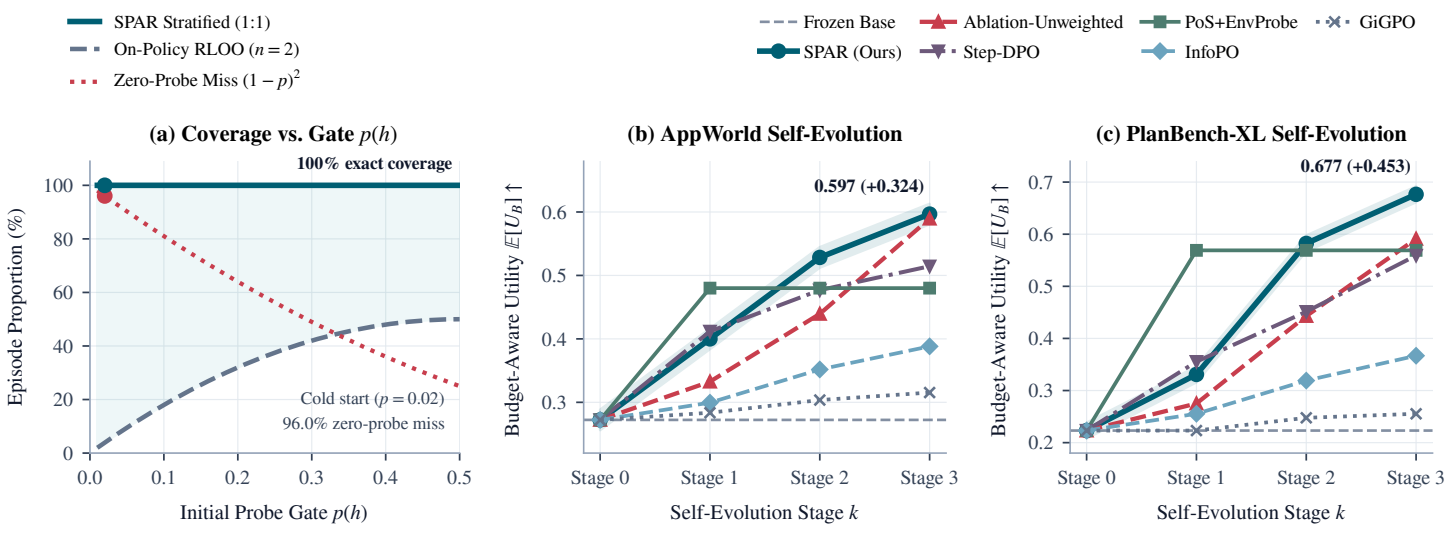
SPAR Stratified (1:1)
On-Policy RLOO ($n = 2$)
Zero-Probe Miss $(1-p)^2$
Frozen Base
Ablation-Unweighted
PoS+EnvProbe
GiGPO
SPAR (Ours)
Step-DPO
InfoPO
(a) Coverage vs. Gate $p(h)$
100% exact coverage
Cold start ($p = 0.02$)
96.0% zero-probe miss
Episode Proportion (%)
Initial Probe Gate $p(h)$
(b) AppWorld Self-Evolution
0.597 (+0.324)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
Stage 0
Stage 1
Stage 2
Stage 3
Self-Evolution Stage k
(c) PlanBench-XL Self-Evolution
0.677 (+0.453)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
Stage 0
Stage 1
Stage 2
Stage 3
Self-Evolution Stage k